

Project Development Phase

User Acceptance Testing (UAT)

Introduction

User Acceptance Testing (UAT) validates that the Metro Ticket Generation System meets business requirements and user expectations. Testing is performed from the perspective of end users such as passengers, administrators, and ticket inspectors.

Objectives

- Verify that business requirements are satisfied.
- Confirm the booking workflow is easy to use.
- Validate fare calculations.
- Verify QR ticket generation.
- Ensure booking notifications are delivered.
- Confirm ticket verification works correctly.

UAT Participants

User Role	Responsibility
Passenger	Book metro tickets
Metro Administrator	Manage bookings and stations
Ticket Inspector	Verify QR tickets
System Administrator	Monitor system configuration

User Acceptance Test Cases

Test ID	Test Case	Expected Result	Status
UAT-01	Book a metro ticket	Booking completed successfully	Pass
UAT-02	Select source and destination	Correct stations displayed	Pass
UAT-03	Choose journey type	Selection accepted	Pass

Test ID	Test Case	Expected Result	Status
UAT-04	Enter passenger count	Value accepted and validated	Pass
UAT-05	Calculate fare	Correct fare displayed	Pass
UAT-06	Generate QR ticket	QR generated successfully	Pass
UAT-07	Receive confirmation email	Email received with ticket details	Pass
UAT-08	View booking history	Booking appears in history	Pass
UAT-09	Scan QR ticket	Ticket verified successfully	Pass
UAT-10	Invalid input validation	Error message displayed	Pass

UAT Checklist

Requirement	Verified
User login	✓
Ticket booking	✓
Journey validation	✓
Fare calculation	✓
Ticket generation	✓
QR generation	✓
Notification delivery	✓
Booking history	✓
QR verification	✓
Admin management	✓

User Feedback

Passenger Feedback

- Ticket booking process is simple.
- QR ticket generation is fast.

- Fare calculation is accurate.
- User interface is easy to understand.

Administrator Feedback

- Booking records are well organized.
- Reports are easy to access.
- Station management is straightforward.

Ticket Inspector Feedback

- QR verification is quick.
- Ticket information is displayed correctly.
- Invalid tickets are identified accurately.

UAT Summary

The Metro Ticket Generation System successfully met the functional and usability requirements defined during the requirement analysis phase. Users were able to complete ticket booking, receive QR-based tickets, and verify them without issues. The system demonstrated ease of use, reliable performance, and accurate data processing, indicating readiness for deployment.